Alpine Swift Migration Dashboard - Technical Report

CCES FS26 - Interactive Environmental Data Dashboard

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Abstract

This report documents an interactive R Shiny dashboard that visualises the trans-Saharan migration of the Alpine Swift (*Tachymarptis melba*). Light-level geolocator tracks for 173 individuals from nine colonies across four populations in Europe are used as the data basis for the dashboard. The dashboard follows the geovisual analytics framework and aims to provide a tool for in-depth exploration of this dataset. Key functionalities include a temporal animation of migration across the annual cycle, linked statistical views, and interactive filtering by population and migration path (flyway)

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1 Dashboard Context and Goals

The Alpine Swift (*Tachymarptis melba*) is a small aerial insectivore that spends the majority of its life airborne. The species is capable of completing trans-Saharan migration in approximately one week, which represents one of the shortest migration durations recorded among long-distance migrants (Meier, Karaardıç, Aymí, et al. 2020). Between 2014 and 2016, the Swiss Ornithological Institute conducted a long-term tracking study of the trans-Saharan migratory cycle using light-level geolocators. Individuals from four populations were tagged along a twelve-degree latitudinal gradient: Pirasali Island (Turkey), Tarragona (Spain), Sofia (Bulgaria), and multiple colonies in Switzerland. The principal finding of Meier, Karaardıç, Aymí, et al. (2020) is that eastern and western populations follow two geographically distinct flyways across the Sahara, with the western populations migrating via the Iberian Peninsula and the eastern populations via Egypt. Furthermore, more southern colonies depart for their wintering grounds later in autumn, while more northern colonies depart from their wintering grounds later in spring, resulting in a breeding season up to 48 days shorter for the northernmost population relative to the southernmost (Meier, Karaardıç, Aymí, et al. 2020). The dashboard presented here provides an interactive analysis tool for exploring these datasets and following the geovisiual analytics approach. As animation is one of the core interactions in geovisual analytics (derived from slides) and therefore comes very naturally as an extention topic and is of great us when it comes to visualize seasonally dependent migration and movement in generall.

2 Data

The dashboard uses data from the long-term tracking project mentioned above, which includes data from 2014 to 2017. It covers 173 Alpine Swifts from the nine different colonies. Each bird carried a light-level geolocator which records the light twice a day and calculates the bird's position from the timing of the sunrise and sunset (Meier, Karaardıç, Aymí, et al. 2020). The tracks for every colony are published as a separate dataset on the Movebank Data Repository (Meier, Peev, et al. 2020; Meier, Karaardıç, and Liechti 2020; Meier, Aymí, et al. 2020; Meier and Liechti 2020a, 2020d, 2020e, 2020b, 2020f, 2020c). It should be noted that there are no positions for the summer breeding months. That's because Alpine Swifts nest in dark cavities in the summer, and every time a bird ducks in or out, the sensor records a "false dusk" or "false dawn" effect which breaks down the position calculation (Lisovski and Hahn 2012). Meier, Karaardıç, Aymí, et al. (2020) stripped these spurious events out with the SGAT package (Lisovski et al. 2020) and calibrated the tags during the stationary November-February window so all devices stay comparable. You'll also notice that the wintering points in Africa sit on a strangely regular grid. That's not a problem of the coordinate reference system. Meier, Karaardıç, Aymí, et al. (2020) processed the raw light data with the FLightR package, which doesn't return a continuous coordinate. It splits space into cells and picks the most likely cell at each time step. During the long stationary winter, when birds stay at almost the same latitude for months, FLightR collapses the daily estimate onto the cell with the highest probability, which is what produces those aligned dots. Two well-known limits of light-level geolocation make this gridding the right call. First, latitude is recovered from day length, and day length is identical at every latitude on Earth around the spring and autumn equinoxes, so for a few weeks each year the latitude estimate is essentially undefined. Meier, Karaardıç, Aymí, et al. (2020) state that "the uncertainty becomes very large, especially if a bird moves predominantly along the same latitude". This is exactly what a swift wintering in a narrow tropical band does. The animated map shows this directly in the migration uncertainty layer, where every migration fix carries a thin vertical "uncertainty shadow" running from the 10th to the 90th percentile of latitude in a 7-day centred rolling window per bird-year, so the diffuse band widens visibly around the equinoxes. Second, the cavity-shading problem already mentioned removes many usable twilights. Collapsing the low-information leftovers onto a stationary grid is, in that context, the honest choice, because it shows the bird's wintering territory rather than a daily walk the device could never actually measure.

2.1 Data preprocessing

Before any data reaches the dashboard, a small pipeline in R/data_processing.R cleans the raw Movebank CSVs and adds the derived columns we need for the visualisation. Each fix is first labelled as breeding, migration or wintering by reading the Movebank comments column, which carries the phase classification published by Meier, Karaardıç, Aymí, et al. (2020). An airspeed filter then drops every pair of consecutive fixes that would require more than 50 km/h average ground speed to remove obvious sensor jitter according to the species cruising speed of roughly 12.6 m/s (45.4 km/h) reported in Meier, Karaardıç, Aymí, et al. (2020). The surviving fixes are collapsed to one daily-median position per bird per date, and for every bird-year a centred 7-day rolling window of the 10th and 90th latitude percentile (lat_lo, lat_hi) is computed. This is the shadow band the migration-uncertainty layer draws on the map. A per-bird phenology summary (arrival, departure, length of stay at the breeding site) is calculated last. The full processed table is cached to data/processed/tracks_processed.rds so the dashboard launches in under a second on subsequent runs.

3 Project structure

The dashboard is a modular R Shiny application. app.R is the entry point and sources seven files from R/: helpers.R (colour palettes, basemap configuration, legend builder), data_acquisition.R (Movebank CSV ingestion with a synthetic fallback), data_processing.R (the pipeline described above), and four UI / server modules — mod_filters.R (sidebar filters), mod_metrics.R (top-right KPI strip), mod_phenology.R (latitude-by-day-of-year plot) and mod_animation.R (integrated map, server-driven animation, click handler and on-map legend). Styling lives in www/custom.css; the raw and processed data sit under data/; the About-tab text, the bibliography and this report itself live under reports/. Cross-module communication is one-directional, flowing through the reactives returned by filters_server(), which keeps each module independently testable and the dependency graph easy to follow.

4 Storytelling Concept and Intended Insights

The aim of geovisual analytics is to detect the expected while enabling the discovery of the unexpected (Thomas and Cook 2005). To support this, the geovisual analytics approach follows Shneiderman's information-seeking mantra, which drescribes first providing an overview of the complete dataset, then offering utilities for zooming and filtering, and finally enabling access to additional details on demand (Shneiderman 1996). The storytelling concept of this dashboard follows this structure. The initial view is intended to convey that seasonal migration occurs between Europe and Africa, and that two primary flyways exist, one via the Iberian Peninsula and one via Egypt. Subsequent filtering and zooming interactions are designed to allow users to observe that higher-latitude colonies depart earlier in autumn and return later in spring. Beyond these primary findings, the dashboard is intended to support open-ended exploration of additional patterns within the data.

To guide the dashboard design, the Cairo Visualisation Wheel was applied as a conceptual framework (Cairo 2012). This tool allows a visualisation designer to evaluate a product along six dimensions and assess whether it tends toward greater complexity and analytical depth or toward greater simplicity and accessibility. A schematic of the wheel is provided in Figure 1. It is argued here that a dashboard intended for geovisual analytics should position itself toward the complex and analytical end of the spectrum. The following design positions were adopted based on this reasoning.

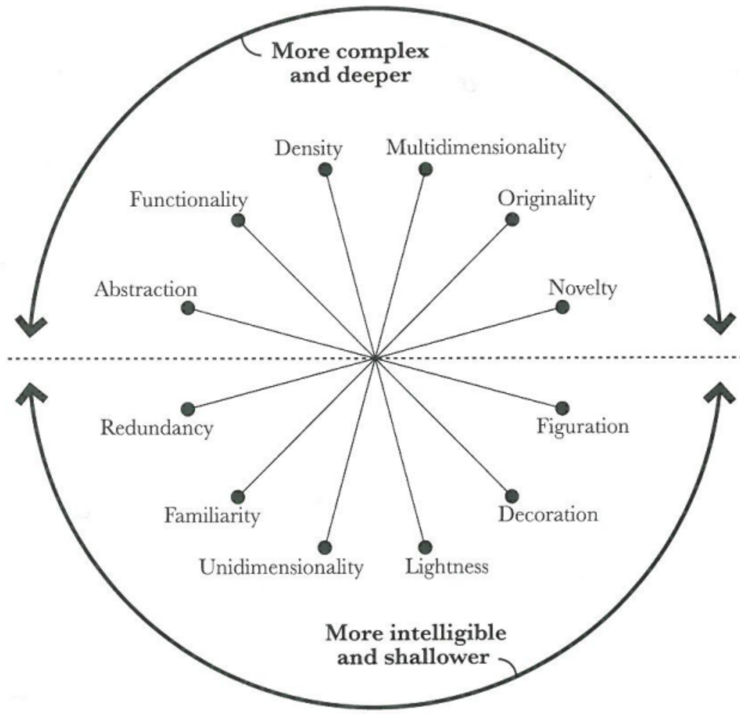

Figure 1: The Visualisation Wheel proposed by Cairo (2012) allows evaluation of a visualisation along six dimensions. Each axis represents a trade-off between two opposing design priorities.

Multidimensionality – Unidimensionality: The dashboard is designed to allow examination of the data from multiple analytical perspectives. Users are able to reclassify observations along the temporal dimension as well as by categorical attributes such as flyway membership or colony of origin.

Density – Lightness: Available screen space is used densely, with a large volume of individual position fixes displayed simultaneously to support pattern recognition across the full dataset.

Functionality – Decoration: The intended audience consists of specialists with an interest in the underlying scientific data. Decorative elements are therefore omitted in favour of an objective and information-dense presentation.

Originality – Familiarity: Familiar visualisation forms, including animated point maps, line charts and boxplots, are preferred over novel representations. Familiar graphics reduce the cognitive overhead required for interpretation and lower the risk of misreading the data. It is worth pointing out that familiar visualisation types push the dashboard towards a more intelligible and shallower visualisation product, which is opposing our argumentation, however we believe that this choice is justified given that this facilitates interpretability.

5 Key Dashboard Functionalities

5.1 Spatial Map

The central element of the dashboard is an interactive map, which serves a dual purpose as both a static data display and an animation canvas.

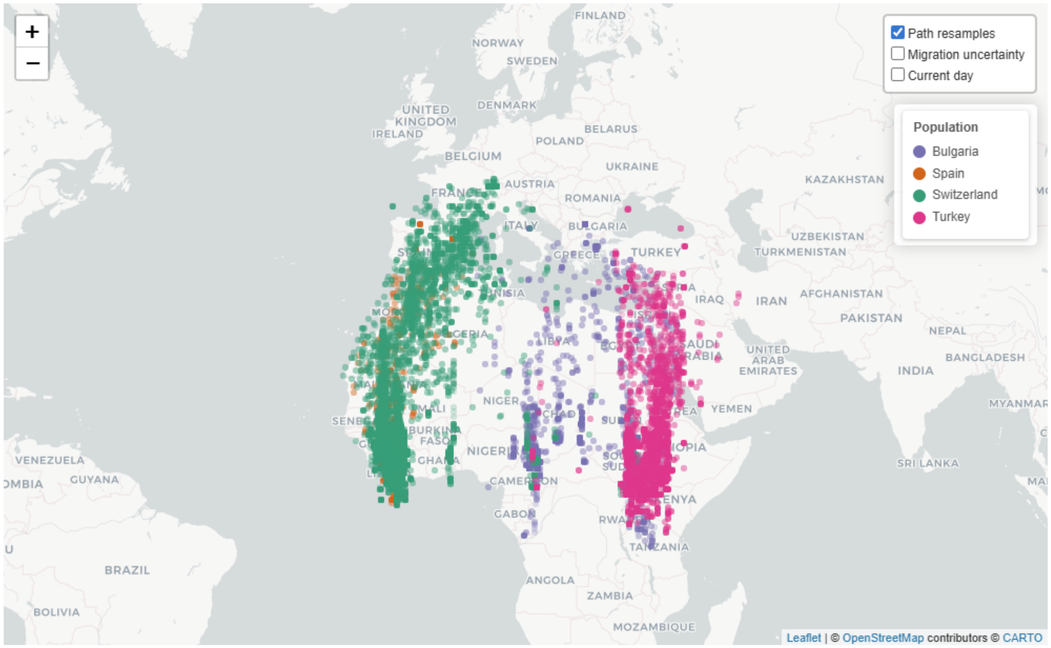

Figure 2: Static view of the spatial map providing a dataset overview upon initial load.

In its default static state, the map displays all retained daily position fixes simultaneously, as shown in Figure 2. This implements the "overview first" principle of Shneiderman's mantra: upon entering the dashboard, the user is immediately presented with a comprehensive spatial overview from which the two flyways and the European–African migration corridor can be identified.

When a temporal animation is active, the map transitions to display the positional development of tracked individuals over the course of the annual cycle, as illustrated in Figure 3.

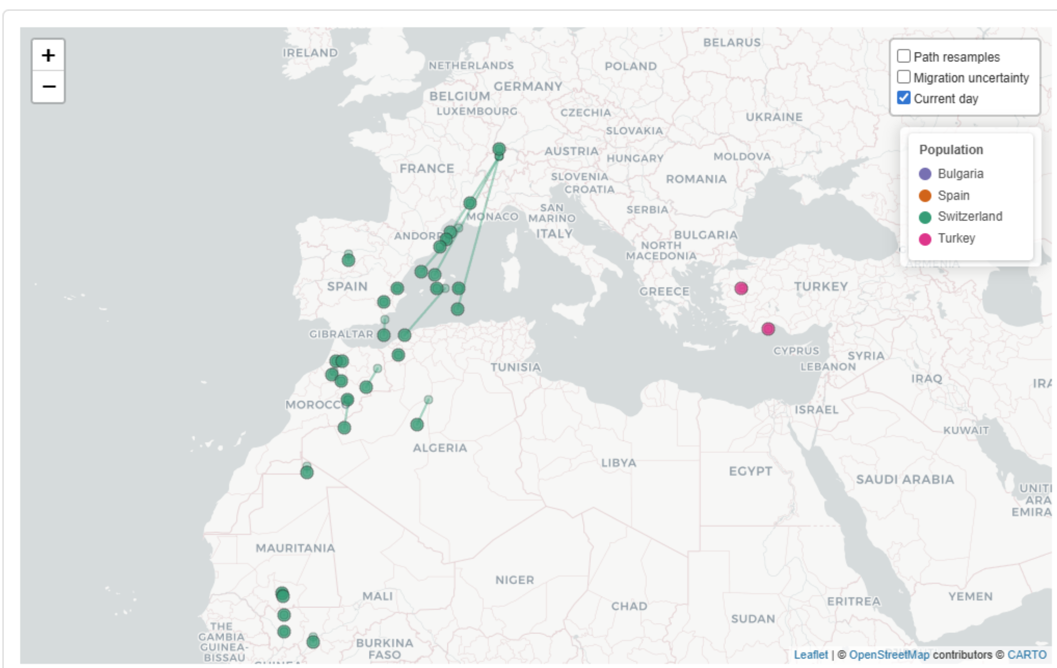

Figure 3: The spatial map during playback of the temporal animation.

5.2 Filter Menu

The filter menu enables restriction of the displayed data by various attributes. Individual colonies can be selected or deselected via a dropdown menu, and a year slider controls which annual tracking periods are included. Pre-configured quick filters are provided for common analytical tasks. In particular, the western and eastern flyways can be selected with a single action. This functionality is designed to guide users toward the primary narrative insight, namely the existence of two geographically distinct migration corridors. An example of the quick filter and year-based colouring in use is shown in Figure 4.

5.3 Attribute Selection Menu

Below the filter menu, a separate control allows users to colour position fixes according to a selected attribute, enabling visual stratification of the data by country of origin, colony, flyway, or year. This attribute selection menue links to cairos dimension of unidimensionality-multidimensionality as it allows a multidimensional view onto the data. Country, colony and flyway are nominal attributes and are therefore encoded using a qualitative ColorBrewer palette. Figure 4 illustrates the map coloured by year membership following application of the western flyway quick filter.

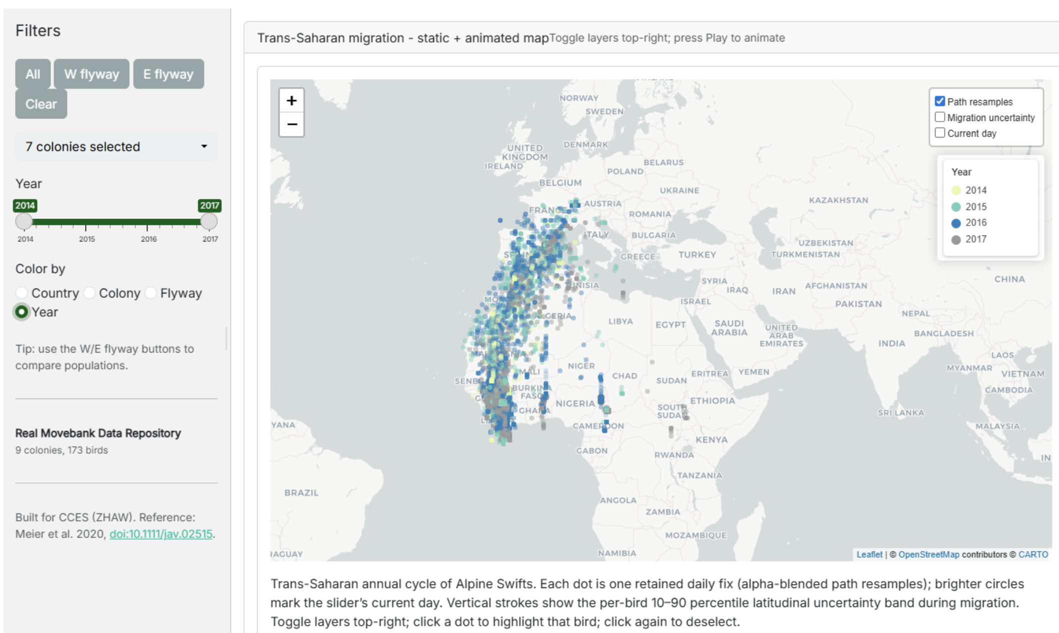

Figure 4: Static map view showing only western flyway datapoints following application of the quick filter. Position fixes are coloured by year of recording using a sequential colour scheme.

5.3.1 Integration into Animation

Both the filter menu and the attribute selection menu remain active during animation playback. Figure 5 provides an example with the western flyway filter active and colouring applied by country of origin.

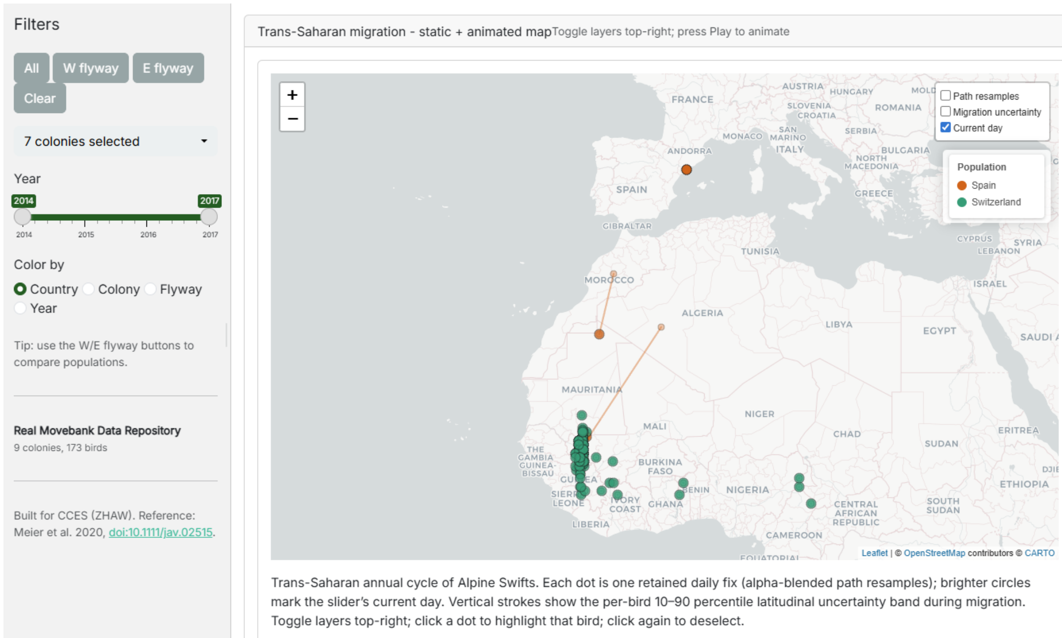

Figure 5: Filter and attribute selection applied during animation. Displayed fixes are restricted to the western flyway and coloured by country of origin.

5.4 Geovisual Analytics Interactions

The dashboard implements several of the canonical geovisual analytics interactions.

Panning, zooming, and navigation are provided through the interactive map, allowing users to adjust the spatial extent and level of detail.

Brushing and mouseover: Hovering over any position fix displays a tooltip containing the individual tag identifier, geographic coordinates, colony membership, and date of recording.

Linked views: Two plots are linked to the spatial map. A set of boxplots displays the current latitudinal distribution of active individuals grouped by the selected attribute, and updates dynamically during animation playback (Figure 6). A static line plot displays the latitudinal position of each tracked individual across the full annual cycle, with lines coloured by the selected grouping attribute (Figure 7). This plot is particularly central to the intended narrative, as it visualises that more northern populations of alpine swift leave europe earlier and depart from africa at a later point, when compared with more southern populations.

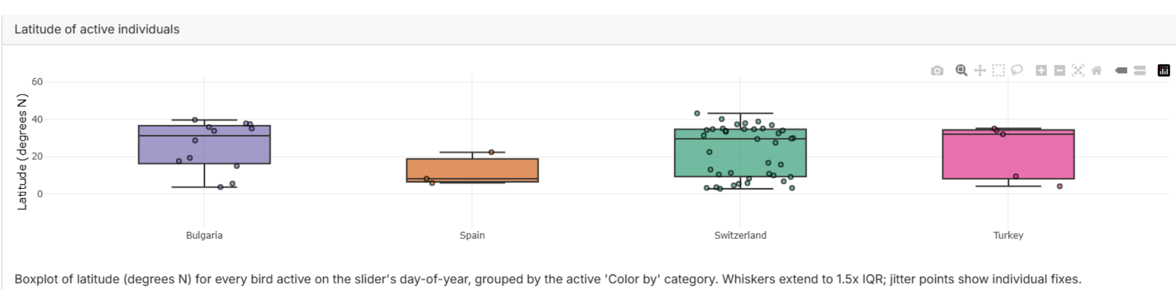

Figure 6: Boxplots displaying the latitudinal distribution of individuals during animation, coloured by country of origin. Individual points represent single tracked birds.

Figure 7: Static line plot displaying the latitudinal position of each tracked individual across the annual cycle. Each line corresponds to one individual. Colouring is applied by country of origin. It is visible that individuals from switzerland leave their wintering ground in africa later and leave their breading ground in europe earlier when compared to spanish individuals.

Both Boxplot as well as line plot are well known and easy to read visualisation types which was specifically choosen to be conform with our decision to lean towards familiarity in cairos Familiarity-Originality dimension.

Animation: The dashboard supports server-driven temporal animation of position fixes across the day-of-year axis. An optional moving-point trail can be toggled, displaying each individual's trajectory over the preceding one, three, or seven days, as visible in Figure 3.

6 Individual Contributions

Lukas Buchmann was responsible for the implementation of the dashboard and reviewed the report. Etienne Duerig implemented the data processing pipeline and authored the report.

7 Declaration of AI Use

The large language models Claude Sonnet 4.6 and Claude Opus 4.5 were used as tools in this project. The models assisted with the generation of dashboard code, including sections produced through a prompt-to-code workflow in which the model was tasked with implementing specific functionality autonomously. Additional code sections were written manually. The models were further employed for code debugging, spelling correction, and formatting of the Quarto report. All generated code was reviewed before integration into the project.

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